DIGITAL LOGIC & DDCA PROJECT

Digital Error Detection Using Parity

Design and simulation of a 4-bit even parity generator and checker using only XOR gates.



Laboratory Implementation

4-Bit Word Architecture · Logisim Verification · Single-Bit Detection

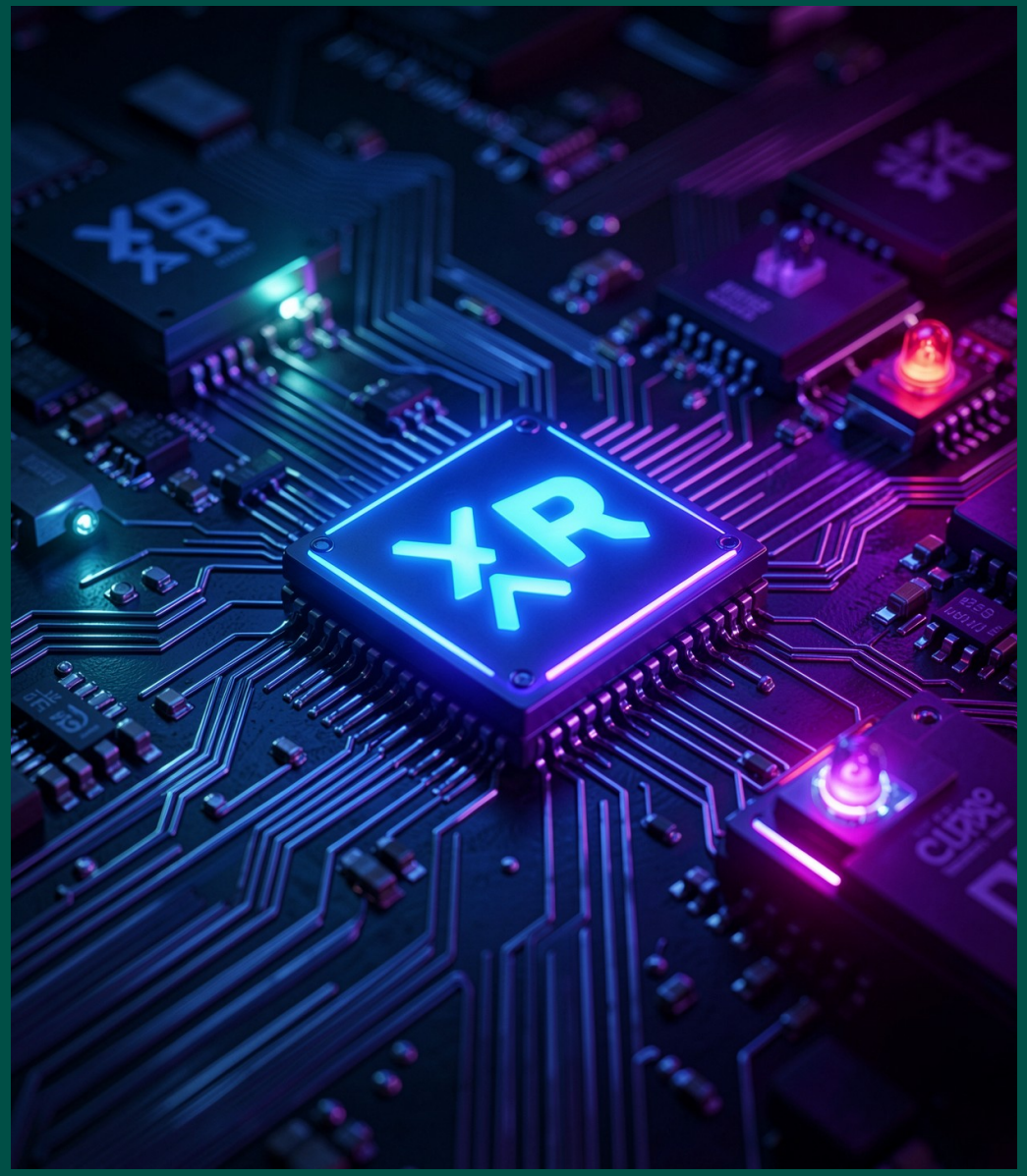
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PROJECT OVERVIEW

Unlocking Reliable Data: Our Parity Detection Project

Parity bits introduce calculated redundancy to data, enabling the detection of single-bit transmission or storage errors in digital systems. This project demonstrates this fundamental principle.

- Design a 4-bit even parity generator using only XOR gates.
- Build a parity checker circuit to flag single-bit errors.
- Verify both circuits exhaustively through Logisim simulation.
- Analyze the detection capabilities and inherent limitations.



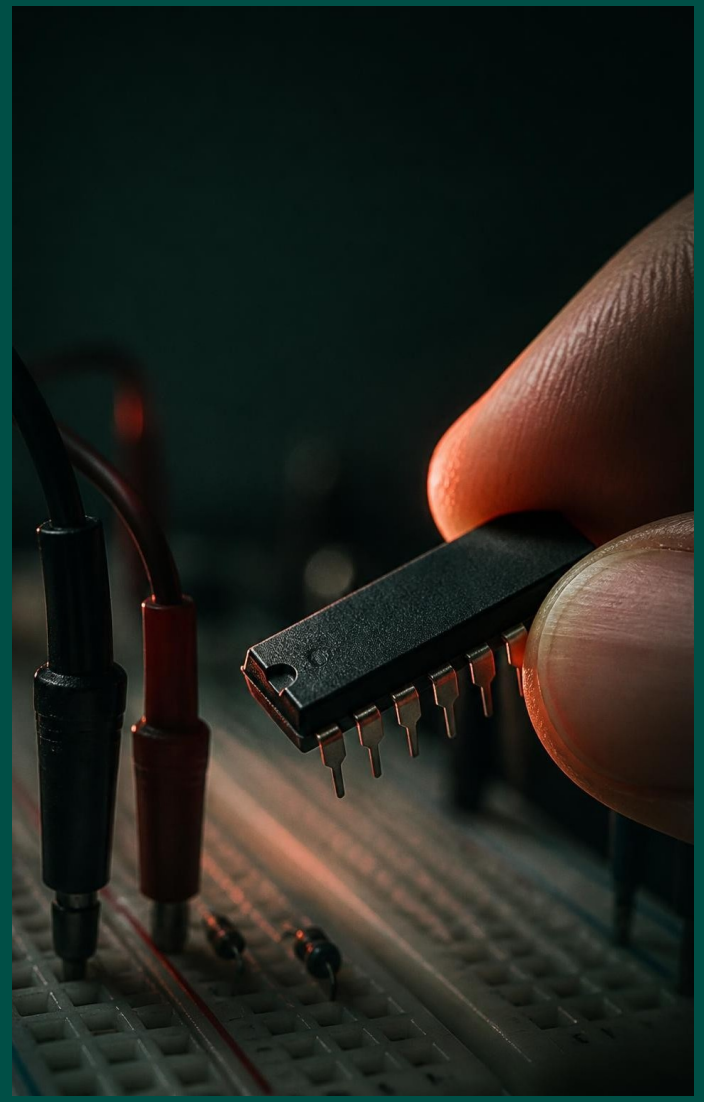
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One extra bit lets the receiver notice a flipped bit

Noise can flip a bit in transit; a parity bit is appended so the receiver can spot it. Even parity keeps the total 1s in data plus parity bit even.

- **Example:** Data 1011 has three 1s, so P = 1 and the sent word 1011 1 has four 1s.
- **XOR property:** XOR outputs 1 for an odd number of 1s, so chained XORs give parity directly.

A	B	Y = A ⊕ B
0	0	0
0	1	1
1	0	1
1	1	0



7486 Quad 2-Input XOR IC architecture

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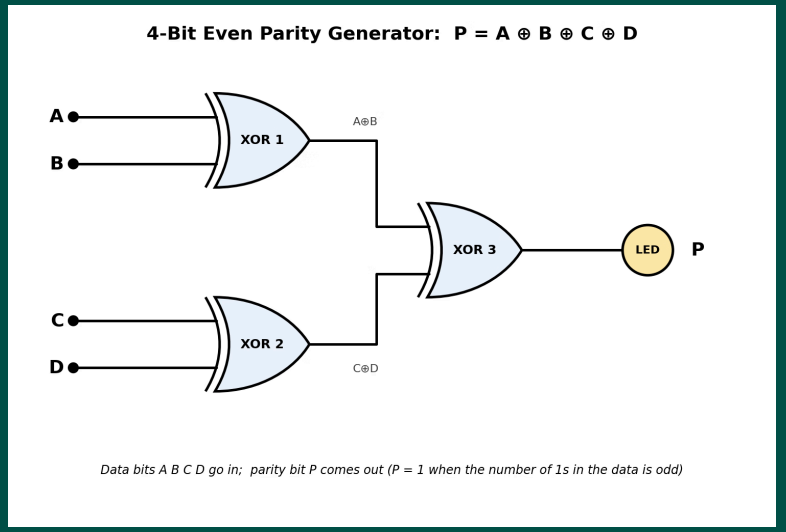
The generator builds P from three XOR gates

$$P = A \oplus B \oplus C \oplus D$$

XOR 1 evaluates A ⊕ B, XOR 2 evaluates C ⊕ D, and XOR 3 combines both outputs to yield parity bit **P**.

Inputs (A B C D)	Parity Bit (P)
0 0 0 0	0
0 0 0 1	1
0 1 1 1	1
1 1 1 1	0

16 combinations in full.



Logisim Parity Generator Circuit: Inputs A, B, C, D to Output P

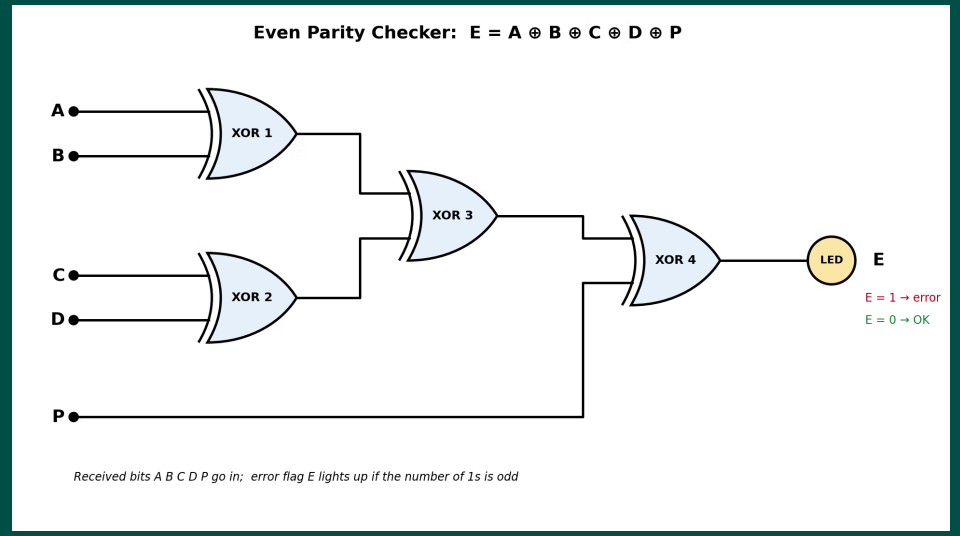
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The checker adds the received bit and flags odd parity

Four XOR gates sum all received data lines plus parity bit P to produce error signal **E**:

- E = 0 indicates even parity; valid codeword with no detected error.
- E = 1 indicates odd parity; an error LED triggers immediately.

Across all 32 possible 5-bit input combinations, exactly **16 combinations** represent invalid codewords and assert the error flag.



Logisim Parity Checker Circuit: Inputs A, B, C, D, P to Error LED E

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Simulation in Logisim confirmed every combination

1

1. Generator Build

Constructed using 3 XOR gates with an LED probe monitoring parity output P.

2

2. Checker Build

Constructed with 4 XOR gates and an error LED E, with dedicated input pins.

3

3. Full State Sweep

Tested all 16 generator inputs and 32 checker vectors; E held 0 for every valid word.

4

4. Fault Injection

Flipping any single bit of the 5-bit transmitted word forced E = 1 without exception.

Complete Hardware Bill: 7 XOR Gates (two 7486 packages) · 5 Input Pins · 2 Probing LEDs

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Simple and low-cost, but detection only

Detects any odd number of flipped bits; misses even numbers because the effects cancel

Cannot locate the wrong bit, so no correction; Hamming code does that

Odd parity: replace the final XOR with an XNOR

Conclusion: a low-cost way to detect single-bit errors in communication and memory systems

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PROJECT OUTCOME

Conclusion and Real-World Applications

Our 4-bit parity system, leveraging 3 XOR gates for generation and 4 for checking, reliably detects any single-bit error in transmission or storage.

Practical Applications

- Computer Memory: Essential for RAM error detection and correction units (ECC memory).
- Serial Communication: Ensures data integrity in links like UART and RS-232 transmissions.
- Storage and Data Buffers: Protects data stored temporarily against transient errors.
- Network Data Transfer: A foundational layer for error checking in various communication protocols.

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